MUHAMMAD FAIZAN RAUF

Solution Architect | AI/ML & Cloud Systems Engineer

Next.js | NestJS | Python | LLMs | Enterprise RAG | AI Agents & Automation | AWS | Docker | CI/CD

Lahore, Pakistan | +92 334 1721711 | faizanrauf6@gmail.com

LinkedIn: linkedin.com/in/m-faizan-rauf | GitHub: github.com/faizanrauf6 | Portfolio: resume-m-faizan-rauf.vercel.app

PROFESSIONAL SUMMARY

Strategic and result-driven **Solution Architect and AI/ML & Cloud Systems Engineer** with **6+ years of expertise** architecting distributed systems, intelligent automation pipelines, and robust cloud infrastructure across FinTech, Healthcare, E-Commerce, and IoT sectors. Expert at bridging the gap between business goals and technical execution, with specialized proficiency in Anthropic API ecosystem design, Multi-Agent orchestration, and scalable microservices. Proven track record of designing high-throughput backends using Next.js, NestJS, and Python while deploying self-healing architectures on AWS and CI/CD for end-to-end delivery.

TECHNICAL SKILLS

Solution Architect: Distributed Systems Design, Microservices, Event-Driven Workflows, API-First Design, REST APIs, GraphQL APIs, System Observability, Role-Based Access Control (RBAC)

AI / ML / LLM Engineering: Anthropic API, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Context Window Optimization, Vector Databases, Scikit-Learn, NumPy, Pandas, Data Engineering

Cloud & DevOps: Amazon Web Services (AWS), EC2, S3, CloudWatch, Docker, Docker Compose, GitHub Actions, CI/CD, Infrastructure Automation, Deployment Orchestration, Linux Server Administration

Backend Development: NestJS, Node.js, Express.js, Python, FastAPI, Django, Flask, C#, ASP.NET, WebSockets, Socket.IO, Pub/Sub Messaging

Frontend Engineering: Next.js, React.js, TypeScript, JavaScript, Tailwind CSS, Material UI

Databases & Persistence: PostgreSQL, MongoDB, MySQL, Amazon DynamoDB, Redis Caching, SQLite

PROFESSIONAL EXPERIENCE

Solution Architect & AI/ML Engineer | PureLogics | Lahore | Jan 2023 - Present
- Design and deliver distributed application features with Next.js, NestJS, TypeScript, and event-driven microservices, separating service responsibilities and data access.
- Build AI orchestration workflows with Anthropic APIs, RAG pipelines, and MCP-connected agents for cross-platform operational automation.
- Build Python data-processing and ML pipelines with NumPy and Pandas for AI-enabled application features.
- Integrate Socket.IO and Modbus to connect cloud applications with commercial HVAC equipment.
- Configure AWS deployments and Docker environments; troubleshoot releases and optimize application performance.

Lead Full Stack Engineer & DevOps | Mahir Company | Lahore | Feb 2022 - Jan 2023
- Led architecture and development of the web platform, administrative console, and MongoDB-backed APIs with product and design teams.
- Built data-preparation workflows and API integrations supporting AI/ML features and operational automation.
- Collaborated with infrastructure teams on AWS services, Docker environments, and release pipelines for web and API services.

Data Automation Systems Executive | NielsenIQ | Lahore | Sep 2020 - Feb 2022
- Designed data-processing and visualization scripts for analytical software; automated ingestion and repetitive data-handling workflows.

Senior Software Engineer | The A Web Technologies | Islamabad | May 2019 - Feb 2020
- Developed web services and MongoDB-backed data layers; implemented unit tests, resolved defects, and improved application performance.

Mobile Application & Systems Engineer | Creativeminds | Islamabad | Feb 2019 - May 2019
- Designed Java and Kotlin mobile applications, modular client components, and backend integrations with MongoDB-based storage.

PROJECTS

HelloCustom | E-Commerce Integration Platform

- Designed multi-tenant, event-driven integration workflows for product personalization and order fulfillment across marketplace services.
- Connected 5 partner APIs: Etsy, Amazon, Printify, Printful, and ICP; supported Shopify integration.
- Implemented transaction handling and retry mechanisms for automated order-processing workflows.

NurseBrain | Healthcare Data Integration Architecture

- Designed read-only FHIR integrations with 2 EHR systems (Epic and Cerner), enabling real-time patient data access and reducing manual entry.

SpeakShift | AI-Powered Video Communication Platform

- Designed real-time video communication workflows with AI transcription, live captions, and multilingual translation.
- Built data routing and WebSocket-based group chat to coordinate communication across connected clients.

SES DLC / Save Energy Systems | Cloud-Connected IoT Platform

- Built AWS EC2-hosted IoT application services using MySQL, JWT authentication, and Apache.
- Integrated Modbus and serial-port communication to connect HVAC equipment with cloud application workflows.

Mahir Company | Consumer Platform & Core API Infrastructure

- Architected the consumer web platform, admin panel, and scalable MongoDB-backed APIs, streamlining data flows across product layers.
- Automated data preparation and API integration for AI/ML workflows; optimized Docker environments and AWS deployment pipelines for continuous releases.

MAWA | Real-Time Workforce Management

- Designed task-management and task-history workflows with multilingual interfaces for distributed workforce collaboration.
- Integrated audio/video calling and persistent chat to connect teams within the application.

WebShield AI | ML-Based Phishing Analysis

- Built a Next.js/NestJS application with a Python ML engine and explainable website-risk analysis.
- Reported offline temporal evaluation on 36,910 URLs: 96.84% precision, 72.87% F1, and 58.41% recall.
- Implemented GitHub Actions CI for tests, builds, PostgreSQL/Redis integration, and Docker isolation checks.

MSQuare Market | Blockchain Architecture & Marketplace

- Designed on-chain and off-chain wallet-service workflows for crypto-asset acquisition and management across connected platforms.
- Built a Next.js admin portal with role-based access control (RBAC) for requests, transactions, platforms, and users.
- Configured Docker Compose, Redis, and local DynamoDB initialization with automated table creation and seed-data loading.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

AWS Certified Cloud Practitioner (CLF-C02) | In Progress

Anthropic Courses: Claude Platform 101, Claude Code 101, Claude 101

Microsoft Office Expert | Microsoft | Sep 2018

EDUCATION

Bachelor of Science in Software Engineering

COMSATS University Islamabad | Feb 2015 - Mar 2019 | GPA: 3.31

Capstone: Wedding automation system using JavaScript, augmented reality (AR), and virtual reality (VR).

HONOURS & LANGUAGES

Award: 1st Position, Open House Project Exhibition, COMSATS Project & Career Expo | Dec 2018

Languages: English, Urdu (Fluent), Punjabi (Fluent)